Depression Risk Prediction Using AI / Machine Learning – 100 Viva Questions and Answers

1. What is depression?

Answer:
Depression is a common mental health disorder characterized by persistent sadness, loss of interest, low energy, negative thoughts, and difficulty performing daily activities.

2. What is depression risk prediction?

Answer:
Depression risk prediction is the process of using data, statistical methods, and machine learning algorithms to identify individuals who may have a higher probability of developing depression.

3. Why is depression risk prediction important?

Answer:
It helps in early identification of mental health problems, allowing timely support, counseling, and preventive interventions.

4. What is Artificial Intelligence (AI)?

Answer:
Artificial Intelligence is a technology that enables computers to perform tasks that normally require human intelligence, such as learning, reasoning, and decision-making.

5. How is AI used in mental health?

Answer:
AI is used for mental health screening, risk prediction, chatbot support, emotion analysis, diagnosis assistance, and treatment recommendations.

6. What is Machine Learning (ML)?

Answer:
Machine Learning is a branch of AI where computers learn patterns from data and make predictions without being explicitly programmed.

7. Why use machine learning for depression prediction?

Answer:
Machine learning can analyze large amounts of psychological, behavioral, and demographic data to detect hidden patterns related to depression risk.

8. What are the main causes of depression?

Answer:
Depression can be caused by genetic factors, biological changes, stressful life events, academic pressure, financial problems, trauma, and social isolation.

9. What are common symptoms of depression?

Answer:
Symptoms include sadness, loss of interest, sleep problems, appetite changes, fatigue, poor concentration, guilt, and suicidal thoughts.

10. What is the difference between sadness and depression?

Answer:
Sadness is a temporary emotional response, while depression is a persistent mental health condition affecting daily life.

PHQ-9 Related Questions

11. What is PHQ-9?

Answer:
PHQ-9 (Patient Health Questionnaire-9) is a standardized questionnaire used to measure depression severity.

12. How many questions are in PHQ-9?

Answer:
PHQ-9 contains 9 questions based on depression symptoms.

13. What is the scoring range of PHQ-9?

Answer:
The total score ranges from 0 to 27.

14. What does a higher PHQ-9 score indicate?

Answer:
A higher score indicates a higher level of depressive symptoms.

15. What are PHQ-9 severity levels?

Answer:
0–4 minimal, 5–9 mild, 10–14 moderate, 15–19 moderately severe, and 20–27 severe depression.

16. Why use PHQ-9 in depression prediction?

Answer:
It provides a validated measurement of depressive symptoms that can be used as a prediction target.

17. Is PHQ-9 a diagnosis tool?

Answer:
No. PHQ-9 is a screening tool and does not replace professional diagnosis.

18. Who developed PHQ-9?

Answer:
PHQ-9 was developed by researchers including Dr. Robert L. Spitzer, Dr. Janet B.W. Williams, and colleagues.

19. What type of data does PHQ-9 generate?

Answer:
It generates numerical psychological assessment data.

20. How is PHQ-9 used in ML models?

Answer:
PHQ-9 scores can be used as labels or features for training prediction models.

DASS-21 Related Questions

21. What is DASS-21?

Answer:
DASS-21 is a psychological assessment tool measuring depression, anxiety, and stress.

22. How many questions are in DASS-21?

Answer:
DASS-21 contains 21 questions.

23. What does DASS stand for?

Answer:
Depression Anxiety Stress Scale.

24. Why use DASS-21 in research?

Answer:
It provides information about multiple psychological conditions together.

25. What are the three components of DASS-21?

Answer:
Depression, anxiety, and stress.

Dataset Questions

26. What is a dataset?

Answer:
A dataset is a collection of organized information used for analysis and machine learning.

27. What type of dataset is used for depression prediction?

Answer:
Psychological survey data, demographic data, academic information, lifestyle data, and behavioral data.

28. What features can be used for depression prediction?

Answer:
Age, gender, academic stress, sleep quality, financial condition, social support, PHQ-9 scores, and DASS scores.

29. Why is data collection important?

Answer:
High-quality data improves model accuracy and reliability.

30. What is labeled data?

Answer:
Data where the correct output or category is already provided.

Data Preprocessing Questions

31. What is data preprocessing?

Answer:
Data preprocessing is the process of cleaning and preparing raw data before training a machine learning model.

32. Why is preprocessing required?

Answer:
It improves data quality and increases model performance.

33. What are missing values?

Answer:
Missing values are empty or unavailable data fields.

34. How are missing values handled?

Answer:
They can be removed or replaced using techniques like mean, median, or mode imputation.

35. What is normalization?

Answer:
Normalization converts values into a common scale.

36. What is feature scaling?

Answer:
Feature scaling ensures different features have similar ranges.

37. What is encoding?

Answer:
Encoding converts categorical information into numerical format.

38. What is data balancing?

Answer:
Data balancing ensures equal representation of different classes.

39. What is SMOTE?

Answer:
SMOTE is a technique used to create synthetic samples for minority classes.

40. Why is balanced data important?

Answer:
It prevents the model from becoming biased toward majority classes.

Machine Learning Algorithms

41. What algorithms can be used for depression prediction?

Answer:
Logistic Regression, Decision Tree, Random Forest, SVM, XGBoost, Neural Networks, and Deep Learning models.

42. What is Logistic Regression?

Answer:
A supervised learning algorithm used for classification problems.

43. What is Decision Tree?

Answer:
A model that makes decisions using a tree-like structure of conditions.

44. What is Random Forest?

Answer:
Random Forest combines multiple decision trees to improve prediction accuracy.

45. What is Support Vector Machine (SVM)?

Answer:
SVM separates data into classes using an optimal decision boundary.

46. What is XGBoost?

Answer:
XGBoost is an advanced boosting algorithm known for high prediction performance.

47. What is Deep Learning?

Answer:
Deep Learning uses artificial neural networks with multiple layers to learn complex patterns.

48. What is an Artificial Neural Network?

Answer:
A computational model inspired by the human brain structure.

49. What is supervised learning?

Answer:
Learning using labeled examples where input and output are known.

50. What is unsupervised learning?

Answer:
Learning patterns from data without predefined labels.

Model Training Questions

51. What is model training?

Answer:
Training is the process where a machine learning model learns patterns from data.

52. What is training data?

Answer:
Data used to teach the machine learning model.

53. What is testing data?

Answer:
Data used to evaluate model performance.

54. What is validation data?

Answer:
Data used to tune and optimize the model.

55. What is overfitting?

Answer:
Overfitting occurs when a model learns training data too closely and performs poorly on new data.

56. What is underfitting?

Answer:
Underfitting occurs when a model fails to learn important patterns.

57. How can overfitting be reduced?

Answer:
Using more data, regularization, dropout, and cross-validation.

58. What is cross-validation?

Answer:
A technique to evaluate model performance using multiple data splits.

59. What is hyperparameter tuning?

Answer:
Finding the best model settings to improve performance.

60. What is feature selection?

Answer:
Selecting the most important variables for prediction.

Evaluation Metrics

61. What is accuracy?

Answer:
Accuracy measures the percentage of correctly predicted cases.

62. What is precision?

Answer:
Precision measures how many predicted positive cases are actually positive.

63. What is recall?

Answer:
Recall measures how many actual positive cases are correctly identified.

64. What is F1-score?

Answer:
F1-score is the balance between precision and recall.

65. What is ROC-AUC?

Answer:
ROC-AUC measures the model's ability to distinguish between classes.

66. What is a confusion matrix?

Answer:
A table showing correct and incorrect predictions.

67. What are True Positives?

Answer:
Correctly predicted positive cases.

68. What are False Positives?

Answer:
Incorrectly predicted positive cases.

69. What are False Negatives?

Answer:
Positive cases missed by the model.

70. Why is recall important in depression prediction?

Answer:
Missing high-risk individuals can have serious consequences, so detecting positive cases is important.

AI Ethics and Security

71. What are ethical issues in AI mental health systems?

Answer:
Privacy, bias, fairness, consent, and misuse of predictions.

72. Why is privacy important?

Answer:
Mental health information is sensitive personal data.

73. How can data privacy be protected?

Answer:
Using encryption, anonymization, and secure storage.

74. What is AI bias?

Answer:
AI bias occurs when predictions unfairly favor or disadvantage certain groups.

75. How can bias be reduced?

Answer:
Using diverse datasets and fairness evaluation methods.

76. What is Explainable AI (XAI)?

Answer:
XAI provides explanations for how AI models make decisions.

77. Why is XAI important?

Answer:
It increases trust and transparency.

78. What is informed consent?

Answer:
Permission given by participants after understanding the research purpose.

79. Should AI replace psychologists?

Answer:
No. AI should support professionals, not replace human judgment.

80. What are limitations of AI depression prediction?

Answer:
Data quality issues, bias, privacy concerns, and difficulty understanding human emotions.

Project Viva Questions

81. What is the aim of your project?

Answer:
The aim is to develop an AI-based system to predict depression risk using psychological and demographic data.

82. Who are the target users?

Answer:
University students, counselors, healthcare providers, and researchers.

83. Why focus on university students?

Answer:
Students experience academic pressure, financial stress, and lifestyle changes.

84. What programming languages can be used?

Answer:
Python is commonly used for machine learning development.

85. What libraries are useful?

Answer:
Scikit-learn, Pandas, NumPy, TensorFlow, PyTorch, and Matplotlib.

86. What is the role of Python in this project?

Answer:
Python is used for data processing, model training, evaluation, and deployment.

87. What is Streamlit?

Answer:
Streamlit is a framework for creating interactive machine learning web applications.

88. What is deployment?

Answer:
Deployment is making the trained model available for real-world use.

89. What improvements can be added in the future?

Answer:
Real-time monitoring, multilingual support, chatbot integration, and larger datasets.

90. How can this system help society?

Answer:
It supports early detection and encourages people to seek professional help.

91. Why combine PHQ-9 and DASS-21?

Answer:
Combining them provides a broader understanding of mental health conditions.

92. What is natural language processing (NLP)?

Answer:
NLP enables computers to understand and analyze human language.

93. How can social media data help prediction?

Answer:
Text patterns and emotional expressions may provide additional indicators.

94. What is a prediction model output?

Answer:
It can provide risk categories such as low, medium, or high risk.

95. What is classification?

Answer:
Classification assigns data into predefined categories.

96. Why evaluate multiple algorithms?

Answer:
Different algorithms may perform differently depending on the dataset.

97. What makes a good depression prediction model?

Answer:
High accuracy, fairness, explainability, privacy protection, and reliability.

98. What is the biggest challenge in this project?

Answer:
Collecting quality mental health data while maintaining privacy and ethical standards.

99. What is the future scope of this research?

Answer:
Developing personalized mental health support systems using advanced AI.

100. What is the conclusion of the project?

Answer:
AI-based depression risk prediction can assist early identification and provide valuable support when combined with professional mental healthcare.